0.13 % of the Weights Carry 99 % of the Loss: Stem-Exempt INT8 Quantization of a Compact Speech Enhancer

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***Abstract*—Post-training quantization (PTQ) of speech enhancement models is normally applied uniformly, and mixed-precision variants assign precision by layer *type* or by a second-order sensitivity proxy. We report a per-layer PTQ study of a 722 K-parameter Bark-band GTCRN denoiser in which the allocation is chosen on a speaker-disjoint development split and then evaluated once, out-of-sample, on the 824-clip VoiceBank-DEMAND test set and on 1 100 MS-SNSD clips. We set out to test whether ranking layers by a *perceptual* criterion (PESQ) differs from ranking them by an MSE-family criterion, and it does not: across the six quantizable layers the two rankings agree at Spearman $\rho = +0.943$ against SI-SDR and $\rho = +1.000$ against mask-MSE, and the two allocations differ by 0.0003 dB. We report this negative result in full, because it says something useful—for this model class, the cheap MSE-based sensitivity proxy is already perceptually adequate. The study's positive finding came from the sensitivity sweep itself. A single 3.8 KB stem convolution—$0.133\,\%$ of the model's weights—carries $98.7\,\%$ of the total INT8 degradation on the held-out test set. Exempting it produces a model that is statistically indistinguishable from FP32 ($\Delta = 0.0014$ dB, $p = 0.25$, $n = 824$) at $3.2\times$ compression, versus a uniform-INT8 loss of 0.109 dB ($p = 3.7\mathrm{e}{-51}$). The exemption is significant in all nine test conditions and replicates on MS-SNSD. Counter-intuitively the mixed model is also *smaller* than uniform INT8 (916.9 vs 950.7 KB), because quantizing every layer causes the runtime to insert QDQ pairs around 143 additional activation tensors whose graph overhead exceeds the stem's 2.8 KB of weight savings. We further show that layer-wise quantization errors are not additive, and document two runtime limits: the band GRU cannot be quantized at all, and INT8 inference is $1.4\times$ *slower* than FP32 on x86.**

***Index Terms*—post-training quantization, mixed precision, speech enhancement, model compression, ONNX Runtime**

I. Introduction

Compact speech enhancement models are deployed under a memory budget, and PTQ is the cheapest way to meet it—no retraining, no labelled calibration targets. The default is uniform INT8. When that costs too much quality, the standard remedies are quantization-aware training (expensive) or mixed precision, where some layers are kept at higher precision.

Mixed precision needs a rule for *which* layers. Two rules dominate. The first assigns by layer **type**—for RNN denoisers on microcontrollers, Rusci *et al.* [1] quantize the recurrent layers to INT8 and keep the rest at FP16. The second assigns by a **second-order sensitivity proxy**: HAWQ [2] and HAWQ-V2 [3] rank layers by top Hessian eigenvalue or average Hessian trace and allocate bits accordingly.

Both proxies are ultimately about reconstruction error. But the objective that matters for speech enhancement is *perceptual*, and perceptual quality weights frequency bands very non-uniformly. That gap motivated this study: **if the perceptual ranking of layers differs from the MSE ranking, allocating by the perceptual one should win at matched size.**

It does not differ, and we report that plainly (Sec. V-A). What the sweep did reveal is a much sharper structural fact (Sec. V-B): the degradation is not spread across layers at all. It is almost entirely attributable to one tiny layer at the very front of the network.

Contributions.

1) **A negative result, reported in full** (Sec. V-A): for this model at INT8 and 6-layer granularity, perceptual and MSE-family sensitivity rankings coincide. The cheap proxy is already adequate; practitioners need not build a PESQ-in-the-loop allocator.
2) **The stem-exemption result** (Sec. V-B): $0.133\,\%$ of the weights carry $98.7\,\%$ of INT8 loss—validated out-of-sample after the allocation was frozen and timestamped.
3) **A size inversion that contradicts the naive model of quantization**: quantizing *more* layers produced a *larger* file, because QDQ graph overhead outgrew the weight savings. This is invisible in any bits-per-weight analysis.
4) **Condition-level robustness and an explicit cross-corpus caveat** (Sec. V-C–V-D).
5) **Evidence that layer-wise quantization error is non-additive** (Sec. V-E), which invalidates the additive-cost assumption behind greedy per-layer allocation search.
6) **Two hard runtime limits** (Sec. V-F) that bound this problem on ONNX Runtime.

II. Related work and positioning

Mixed-precision PTQ for speech enhancement. The closest prior work is Rusci *et al.* [1], who deploy an RNN denoiser on a 9-core RISC-V MCU using mixed FP16–INT8 PTQ, reporting 1.4–$1.7\times$ memory saving at 0.06 PESQ degradation against 0.3 for uniform INT8. Two differences matter. First, their allocation is by **layer type**—recurrent layers to INT8—whereas ours is derived from measured per-layer sensitivity. Second, and more striking, our result runs in the *opposite direction on the same architectural family*: on our runtime the recurrent layer cannot be quantized at all (Sec. V-F), and the

TABLE I
SPEAKER-DISJOINT SPLITS. THE TEST AND EXTERNAL SETS WERE EACH TOUCHED EXACTLY ONCE, AFTER THE ALLOCATION WAS FROZEN.

split	speakers	clips	used for
calibration	11 train spk.	64	QDQ activation calibration
dev	p254, p256	200	ranking + allocation search
test	p232, p257	824	final numbers
external	MS-SNSD	1100	generalization

layer that must be protected is the small input convolution. We read this as evidence that "which layer to protect" is a property of the *runtime and model*, not a transferable rule—which is an argument for measuring rather than assuming.

Hessian-guided allocation. HAWQ [2] and HAWQ-V2 [3] rank layers by curvature and are the standard sensitivity-based allocators. Our Sec. V-A result is adjacent but distinct: we do not compare against a Hessian proxy; we compare a *perceptual* ranking against *direct output-error* rankings, and find they coincide. Adding a Hessian-trace ranking as a third column is the single most valuable extension to this work.

Speech enhancement compression generally. Other routes to the same budget include sign-exponent-only floating point [5], matrix-product-operator factorisation [6], and structured sparsity for tiny accelerators. These are complementary to PTQ rather than alternatives to it; our contribution is about *where* the loss lives under PTQ, not about beating these methods.

Objective mismatch. The framing—that an MSE-optimal quantizer may be wrong for the objective one actually cares about—was taken from TurboQuant [4]. Its *algorithm* does not apply here: there is no KV cache, no vector database, no memory bottleneck of the kind it addresses. We cite it for motivation only, and the perceptual-vs-MSE form of the hypothesis it suggested was refuted (Sec. V-A).

III. METHOD

Model. A Bark-band GTCRN, `ch96_h96_b4`, 722 209 parameters—a stem convolution, four gated time–frequency convolution blocks with dilations $1/2/4/8$, a per-band GRU decoder, and a linear head producing per-band gains. FP32 size 2842.7 KB.

Quantization. ONNX Runtime `quantize_static`, QDQ format, per-layer control via `nodes_to_quantize`. Activation calibration uses 64 clips drawn from 11 training speakers. Per-layer control was verified to produce genuinely distinct graphs (uniform INT8 $= 172$ QDQ nodes, head-only $= 5$, stem-only $= 6$) with distinct outputs, so the sensitivity study is mechanically meaningful rather than a no-op.

Leakage discipline. Bit allocation is a model-selection decision, so it is made on a development set carved from TRAIN **by speaker**, never on the test set (Table I). The split builder hard-fails on speaker overlap, and the chosen allocation was written to disk *before* any test-set evaluation was run.

Search space. The band GRU is unquantizable on this runtime (Sec. V-F) and INT4 is unavailable for Conv, so the precision axis is per-layer $\in \{\text{FP32, INT8}\}$ over **6** quantizable layers (5 Conv + 1 MatMul)—$2^6 = 64$ configurations, small enough to enumerate exhaustively rather than search greedily. This matters given the non-additivity in Sec. V-E.

TABLE II
RANK CORRELATION BETWEEN SENSITIVITY ORDERINGS OVER THE SIX QUANTIZABLE LAYERS. THE PERCEPTUAL AND MSE-FAMILY RANKINGS AGREE ALMOST PERFECTLY.

ranking comparison	Spearman ρ
PESQ vs SI-SDR	$+0.943$
PESQ vs mask-MSE	$\mathbf{+1.000}$

TABLE III
POST-TRAINING QUANTIZATION ON THE VOICEBANK-DEMAND TEST SET (824 CLIPS, HELD OUT FROM THE DEV SPLIT USED TO CHOOSE THE ALLOCATION). EXEMPTING THE 3.8 KB STEM CONVOLUTION RECOVERS ESSENTIALLY ALL OF THE QUANTIZATION LOSS AND YIELDS A MODEL STATISTICALLY INDISTINGUISHABLE FROM FP32.

Method	Size (KB)	SI-SDR (dB)	vs FP32
FP32 (no quantization)	2842.7	18.377	—
Mixed: stem FP32 (ours)	**916.9**	**18.375**	$p = 0.25$ (n.s.)
Uniform INT8	950.7	18.267	$p = 3.7\text{e}{-51}$

IV. EXPERIMENTAL SETUP

VoiceBank-DEMAND 16 kHz; 824-clip test set with official condition labels (5 noise types $\times$ 4 input SNRs). MS-SNSD provides 1 100 external clips. Metrics: SI-SDR, STOI, ESTOI, LSD, and PESQ. Significance throughout is paired Wilcoxon signed-rank with $95\,\%$ bootstrap CIs on per-clip differences.

V. RESULTS

A. Perceptual and MSE sensitivity rankings coincide (negative result)

`stem` ranks most sensitive under *every* metric; `head` ranks least sensitive under *every* metric (Table II). The two induced allocations differ at exactly one position and produce a ΔSI-SDR of $+0.0003$ dB—indistinguishable.

Conclusion: at INT8 and this granularity, MSE-guided allocation is already perceptually adequate for this model. We make no perceptual-advantage claim.

B. The stem exemption (main result, out-of-sample)

From Table III:

- Stem exemption vs uniform: $+0.108$ dB SI-SDR, $95\,\%$ CI $[+0.091, +0.126]$, $p = 9.2\text{e}{-49}$.
- Mixed vs FP32: $\Delta = 0.0014$ dB, CI $[-0.0012, +0.0041]$, **not significant**.
- Total quantization cost (FP32 $-$ uniform) is 0.109 dB; the stem accounts for $\mathbf{98.7}\,\%$ of it.
- The stem is $\mathbf{3\,840}$ bytes $= 0.133\,\%$ of model weights.

The mixed model is also smaller than uniform INT8. Verified cause: requesting quantization of every layer makes ONNX Runtime wrap $\sim$143 additional activation tensors in QDQ pairs (**172 vs 29** QDQ nodes), and that graph overhead

TABLE IV
MEAN ABSOLUTE PER-CLIP DEVIATION FROM FP32 (LOWER IS MORE FAITHFUL). ALL EIGHT CELLS FAVOUR THE MIXED ALLOCATION.

corpus	metric	mixed	uniform	adv.
VB-824	$\lvert\Delta$SI-SDR$\rvert$	**0.0202**	0.1458	7.2×
VB-824	$\lvert\Delta$STOI$\rvert$	**0.00025**	0.00186	7.4×
VB-824	$\lvert\Delta$ESTOI$\rvert$	**0.00055**	0.00481	8.7×
VB-824	$\lvert\Delta$LSD$\rvert$	**0.0088**	0.1708	19.5×
MS-SNSD	$\lvert\Delta$SI-SDR$\rvert$	**0.0328**	0.2832	8.6×
MS-SNSD	$\lvert\Delta$STOI$\rvert$	**0.00051**	0.00710	13.9×
MS-SNSD	$\lvert\Delta$ESTOI$\rvert$	**0.00079**	0.00966	12.2×
MS-SNSD	$\lvert\Delta$LSD$\rvert$	**0.0209**	0.2915	13.9×

TABLE V
BENEFIT OF EXEMPTING THE STEM, BY NOISE TYPE AND INPUT SNR. THE EFFECT IS STATISTICALLY SIGNIFICANT IN EVERY ONE OF THE NINE CONDITIONS.

Condition	n	ΔSI-SDR (dB)	95 % CI
bus	164	+0.087	[+0.051, +0.130]
cafe	164	+0.156	[+0.124, +0.189]
living	166	+0.098	[+0.064, +0.134]
office	164	+0.024	[+0.011, +0.038]
psquare	166	+0.175	[+0.126, +0.235]
2.5 dB	207	+0.125	[+0.088, +0.165]
7.5 dB	206	+0.095	[+0.065, +0.128]
12.5 dB	207	+0.104	[+0.076, +0.137]
17.5 dB	204	+0.108	[+0.075, +0.151]

exceeds the stem's 2.8 KB of weight savings. Quantizing the stem *alone* even grows the file above FP32 (2851.9 vs 2842.7 KB). Any analysis that counts only bits-per-weight will miss this entirely.

Fidelity to FP32—mean absolute per-clip deviation, the property PTQ is actually supposed to deliver—favours the mixed allocation in **8/8 metric–corpus cells, by** 7–20× (Table IV).

C. Condition robustness

The effect is **flat across SNR** (Table V)—so it is not an SNR artifact—and largest in hard non-stationary noise (psquare, cafe), smallest in easy stationary office noise. That ordering is what one would predict if the stem's early feature extraction matters most when spectral structure is complex, which is a satisfying but *post-hoc* explanation; we offer it as interpretation, not evidence.

D. Cross-corpus caveat—stated explicitly

On MS-SNSD the stem-exemption SI-SDR benefit replicates (+0.039 dB, $p = 3.0\text{e}{-12}$) and mixed remains indistinguishable from FP32 on SI-SDR ($p = 0.17$). **However, on MS-SNSD uniform INT8 scores better than mixed on STOI (−0.0060), ESTOI (−0.0067) and LSD (+0.233).**

This is not evidence that uniform is the better allocation. Uniform INT8 also beats **FP32 itself** on MS-SNSD STOI (0.8212 vs 0.8154)—a quantizer cannot improve on the model it is approximating. Its larger quantization noise acts as a mild dither that flatters intelligibility metrics under distribution shift while degrading waveform fidelity. The mixed model instead tracks FP32 closely on every metric (Table IV).

TABLE VI
LAYER-WISE DEGRADATIONS DO NOT SUM. THE INTERACTION TERM'S CI EXCLUDES ZERO ON BOTH CORPORA.

corpus	stem	5 blocks	combined	ratio
dev (200)	−0.0061	−0.0023	−0.0996	**11.95**×
MS-SNSD (1100)	+0.0260	+0.0010	−0.0377	1.40×

The claim we make is therefore fidelity to FP32, not "better enhancement on every metric everywhere." We state this in the abstract as well as here, because the temptation to overclaim is exactly what this result guards against.

E. Quantization error is not additive

Error **compounds** through the network rather than accumulating independently (Table VI). The magnitude is strongly corpus-dependent, so we report the *existence* of the interaction as robust and its *size* as corpus-specific.

Practical consequence: greedy per-layer allocation search assumes additive costs and is therefore unsound here. Exhaustive enumeration over the 64 configurations was affordable and is what we did.

F. Runtime limits (findings, not bugs)

- **The band GRU cannot be quantized** by ONNX Runtime 1.20.1. Requesting it yields **0 QDQ nodes**, no size change (2852.9 vs 2842.7 KB) and no speedup (0.995×). The recurrent decoder is incompressible on this runtime and is pinned to FP32 in every allocation. Note the direct contrast with [1], whose method depends on quantizing exactly this layer type.
- **INT4 is unavailable for Conv** via `quantize_static`. The runtime's 4-bit path covers MatMul only, and this graph has one tiny MatMul, so INT4 cannot reach the weight-heavy Conv layers.
- **INT8 is ∼1.4× slower than FP32** on x86, re-measured on a verified-idle machine (6 % CPU load, AMD Ryzen 5 5600H, same runtime version): 1.39× / 1.43× / 1.46× at 1 / 2 / 4 threads, with absolute RTF 0.084 (FP32) vs 0.117 (INT8) single-threaded. **On this hardware the benefit of quantization is memory, not speed**—and the disadvantage grows slightly with thread count, consistent with QDQ overhead parallelising worse than the dense kernels it replaces.

VI. LIMITATIONS

The negative result in Sec. V-A is scoped to this checkpoint, at INT8, over six layers. With a finer precision axis (INT4 available, or per-channel), or a model whose layers are less dominated by a single one, the perceptual and MSE rankings could plausibly diverge. We did not show that they never diverge; we showed they do not here, and explained why the search space was forced to be this coarse (Sec. V-F).

One checkpoint, one architecture. The stem-exemption magnitude is not claimed to transfer. The transferable claim is methodological: measure per-layer sensitivity, because the answer is runtime- and model-specific and contradicted a published rule for the same architecture family.

No Hessian baseline. Without it, Sec. V-A compares a perceptual ranking against direct output-error rankings, not against the standard curvature proxy.

No on-device measurement. All timings are x86 with thread pinning as an edge proxy. The contrast with [1] is therefore across both method *and* hardware, and we cannot separate those two factors.

VII. Conclusion

We tested whether perceptually-guided bit allocation beats MSE-guided allocation for a compact speech enhancer, and found that it does not—the rankings coincide almost perfectly. Reporting that negative result is worthwhile on its own: it licenses practitioners to keep using the cheap proxy.

The sweep that produced it revealed something more useful. Quantization loss in this model is not distributed across layers; **$0.133\,\%$ of the weights carry $98.7\,\%$ of it**. Protecting that one 3.8 KB layer yields an FP32-indistinguishable model at $3.2\times$ compression, significant in every test condition, replicated on a second corpus—and, because of QDQ graph overhead, *smaller* than quantizing everything. Combined with the non-additivity of layer-wise errors and the fact that a published mixed-precision rule for this architecture family is inverted on our runtime, the practical lesson is consistent: for PTQ of small speech models, per-layer sensitivity is worth measuring rather than assuming.

References

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